1 Trapezoidal Rule

For a strictly-increasing function f on $[a,b]$, the left-endpoint approximation is always an underestimate of $\int_a^b f(x)dx$, while the right-endpoint approximation is always an overestimate. To get a better estimate of $\int_a^b f(x)dx$, it makes sense to average these two approximations: [1]

$$\begin{aligned}\int_a^b f(x)dx &\approx \frac{1}{2}\left(\sum_{i=1}^{n} f(x_{i-1})\Delta x + \sum_{i=1}^{n} f(x_i)\Delta x\right)\\ &= \sum_{i=1}^{n}\left(\frac{f(x_{i-1})+f(x_i)}{2}\right)\Delta x\\ &= \frac{\Delta x}{2}\left[(f(x_0)+f(x_1))+(f(x_1)+f(x_2))+\cdots+(f(x_{n-2})+f(x_{n-1}))+(f(x_{n-1})+f(x_n))\right]\\ &= \frac{\Delta x}{2}(f(x_0)+2f(x_1)+\cdots+2f(x_{i-1})+f(x_i)).\end{aligned}$$

This approximation is called the *Trapezoidal Approximation*. The reason for this name can be seen from the figure below, which illustrates the case with $f(x)\geq 0$ and $n=4$. The area of the trapezoid that lies above the ith subinterval is

$$\Delta x\left(\frac{f(x_{i-1})+f(x_i)}{2}\right),$$

and if we add the areas of all these trapezoids, we get the right side of the Trapezoidal Rule.

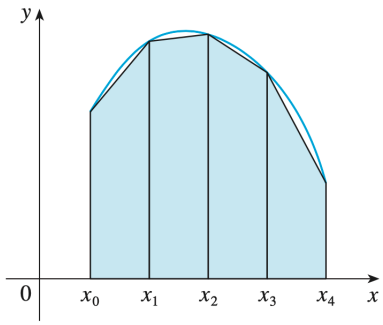


Figure 1: Illustration of the trapezoidal approximation of $\int_a^b f(x)dx$.

[1] As we have seen, the midpoint approximation will also give a better estimate in these situations. We will compare these two approximations in Section 3.

1. Use the Trapezoidal Rule with $n = 5$ to approximate the integral $\int_1^2 \frac{1}{x} dx$ to six decimal places. Compare to the exact answer $\ln(2) \approx 0.693147$ and the midpoint approximation (which you calculated last time) ≈ 0.691908. Which approximation is better?

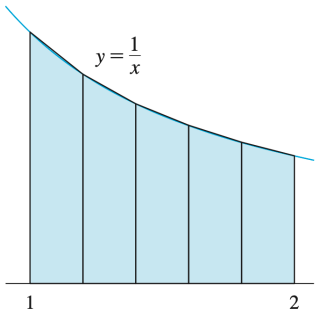


FIGURE 3

EXAMPLE 1 Use (a) the Trapezoidal Rule and (b) the Midpoint Rule with $n = 5$ to approximate the integral $\int_1^2 (1/x)\, dx$.

SOLUTION

(a) With $n = 5$, $a = 1$, and $b = 2$, we have $\Delta x = (2 - 1)/5 = 0.2$, and so the Trapezoidal Rule gives

$$\int_1^2 \frac{1}{x}\, dx \approx T_5 = \frac{0.2}{2}[\, f(1) + 2\, f(1.2) + 2\, f(1.4) + 2\, f(1.6) + 2\, f(1.8) + f(2)]$$

$$= 0.1\left(\frac{1}{1} + \frac{2}{1.2} + \frac{2}{1.4} + \frac{2}{1.6} + \frac{2}{1.8} + \frac{1}{2}\right)$$

$$\approx 0.695635$$

This approximation is illustrated in Figure 3.

2 Error Bound for Trapezoidal Rule

As before, if we don't know the exact value for $\int_a^b f(x)dx$ and we want to approximate it using the Trapezoidal Rule, then we need an estimate of the error in our approximation.

Theorem 2.1 (Error bound for the trapezoidal approximation)**.** *Let E_n^T denote the absolute error $|\sum_{i=1}^n f(x_i^*)\Delta x - \int_a^b f(x)dx|$ using the trapezoidal approximation. If $M = \max\{|f''(x)|\}_{x\in[a,b]}$, then $E_n^T \leq \frac{K(b-a)^3}{12n^2}$.*

We will again postpone the proof so that we can get some practice calculating error bounds.

4. Use the error estimate for the trapezoidal rule to find an upper bound for the error in your approximation in Problem 1. Compare this to the exact error.

5. How large should we take n in order to guarantee that the trapezoidal approximation for $\int_1^2 \frac{1}{x}dx$ is accurate to within 0.0001?

3 Trapezoidal Rule vs Midpoint Rule

Comparing the error bounds we derived for the trapezoidal rule and the midpoint rule, we found that

$$\max(E_n^M) = \frac{(b-a)^3}{24n^2} = \frac{1}{2}\frac{(b-a)^3}{12n^2} = \frac{1}{2}\max(E_n^T). \tag{3.1}$$

These are just upper bounds for the error, not the actual error, so Equation (3.1) does *not* say that the error in the midpoint approximation is half that of the trapezoidal approximation. However, in many cases, the midpoint approximation is more accurate than the trapezoidal approximation. We now give an argument for why this is the case.

6. (Optional, and potentially quite time-consuming) Prove that for a cubic polynomial on $[a,b]$, the errors (note that these are the errors, not the absolute errors) satisfy the equation $E_n^M = -\frac{1}{2}E_n^T$. (*Hint*: Calculate the exact error in each approximation for monomials x^i, $i = 0,\ldots,3$. Then use the fact that the error E satisfies $E(\sum_{i=1}^n c_i x^i) = \sum_{i=1}^n c_i E(x^i)$ for any constant c.) It follows that, if the function f you are integrating is well-approximated by a cubic polynomial on every subinterval in your partition (which, as we will see later, will be the case if f is three-times differentiable), then the errors should still be related in roughly the same way.